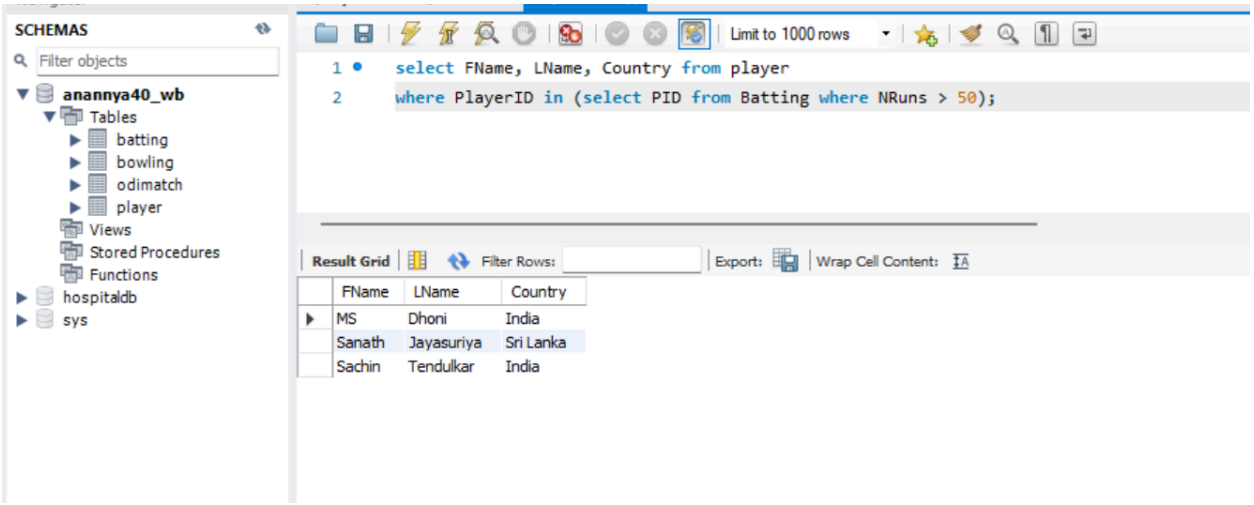
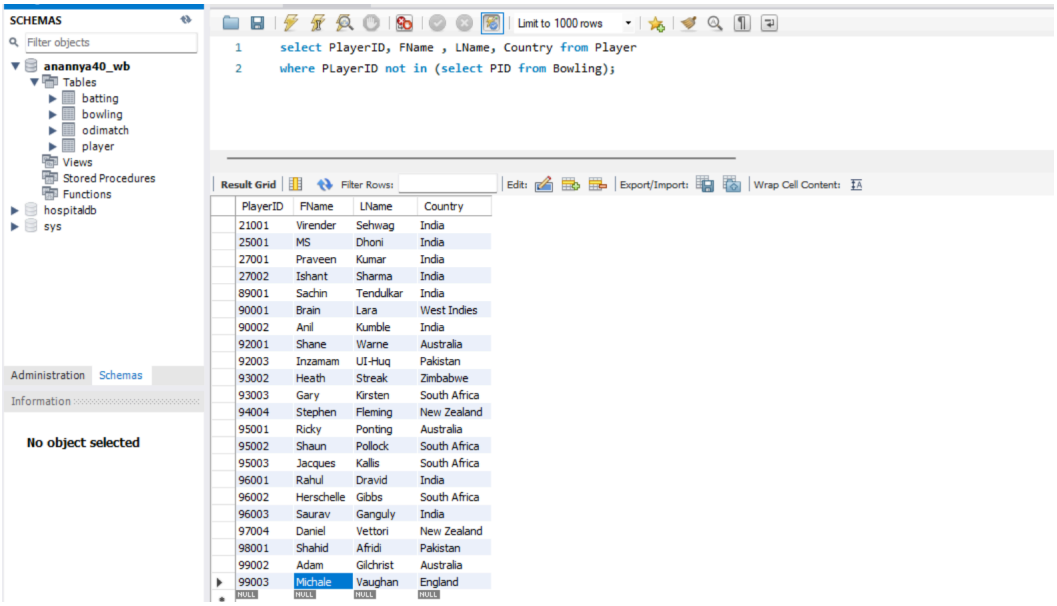
Experiment 7: Perform Nested and Complex Queries

Aim: Mastering Subqueries (IN, NOT IN, EXISTS, NOT EXISTS, ANY, ALL), Correlated Subqueries, Derived Tables (FROM clause), HAVING clauses, and Set Operators (UNION, INTERSECT).

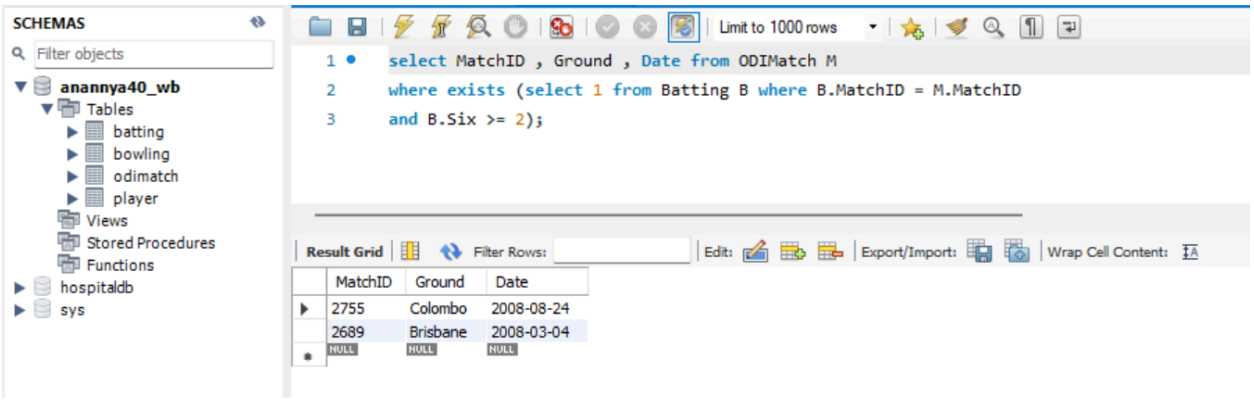
1. [Subquery - IN]: Write an SQL query using an IN subquery to find the full names and countries of players who scored more than 50 runs (NRuns > 50) in any single match.



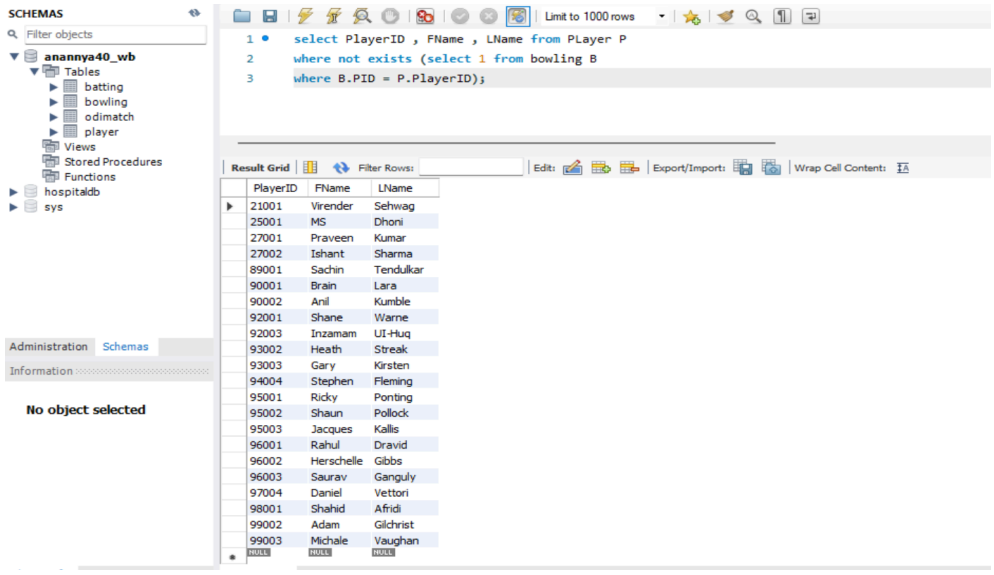
2. [Subquery - NOT IN]: Retrieve all players (PlayerID, FName, LName, Country) who have NEVER bowled in any match using a NOT IN subquery referencing the Bowling table.



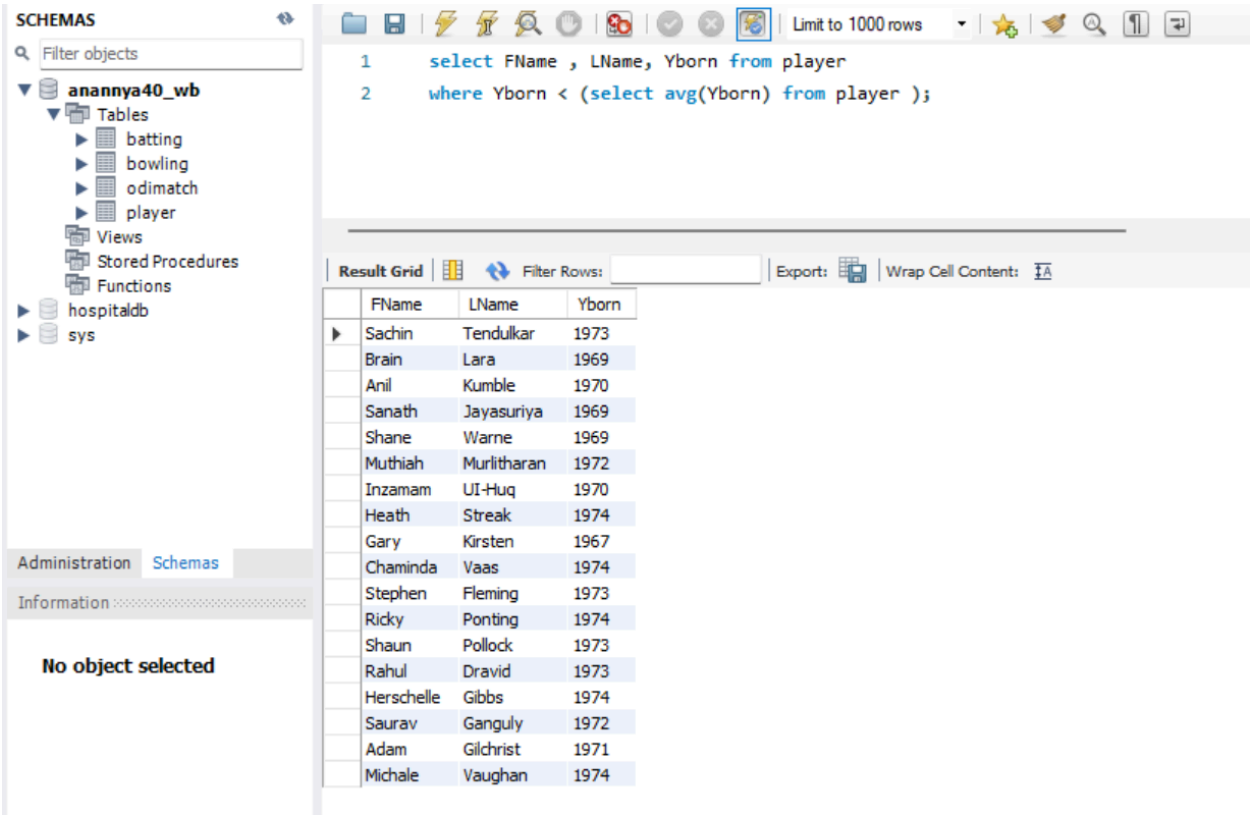
3 .[Subquery - EXISTS]: Write an SQL query using EXISTS to list all matches (MatchID, Ground, Date) where at least one player hit 2 or more sixes (Six >= 2).



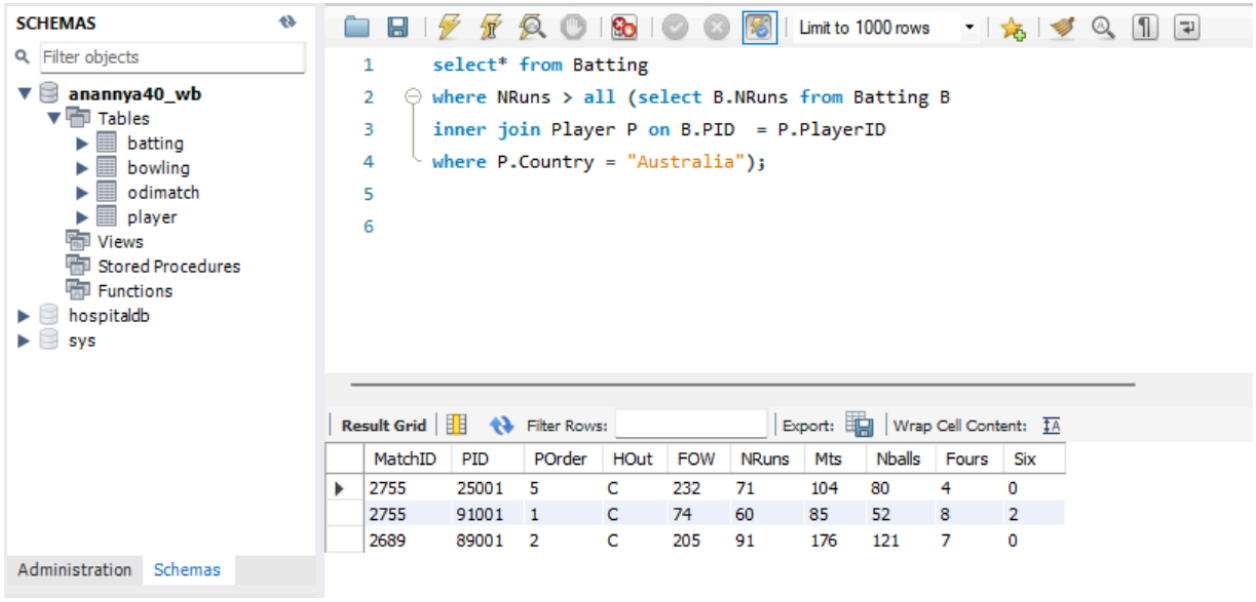
4 .[Subquery - NOT EXISTS]: Retrieve all players (PlayerID, FName, LName) for whom NO bowling record exists in the Bowling table using a correlated NOT EXISTS clause.



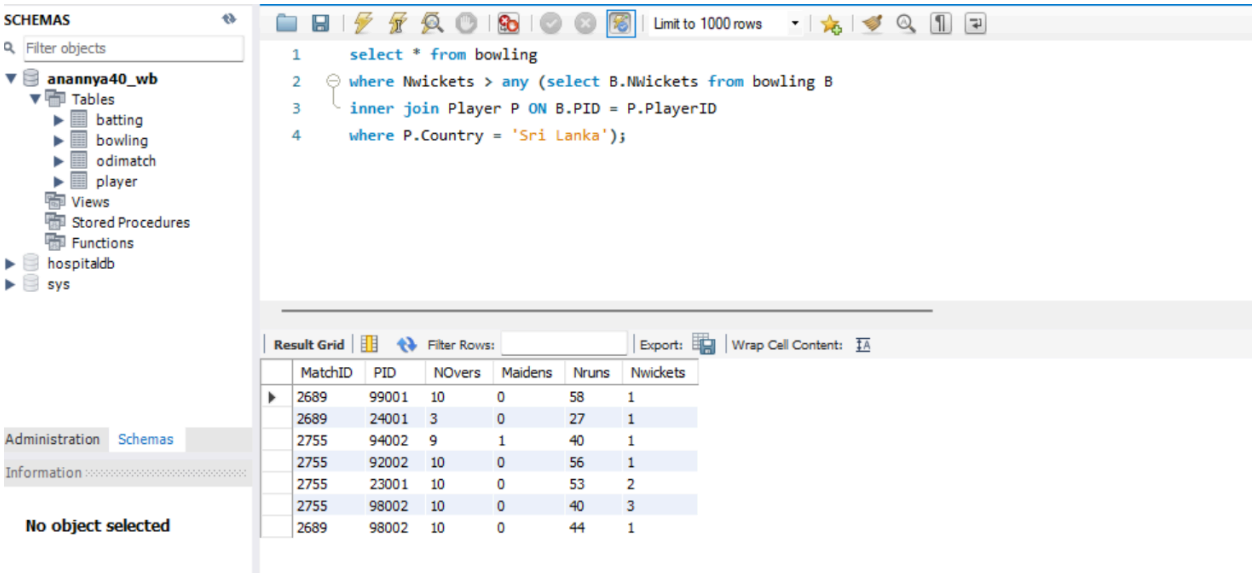
5. [Scalar Subquery in WHERE]: Find all players (FName, LName, Yborn) who were born before the average birth year (Yborn < AVG(Yborn)) of all registered players.



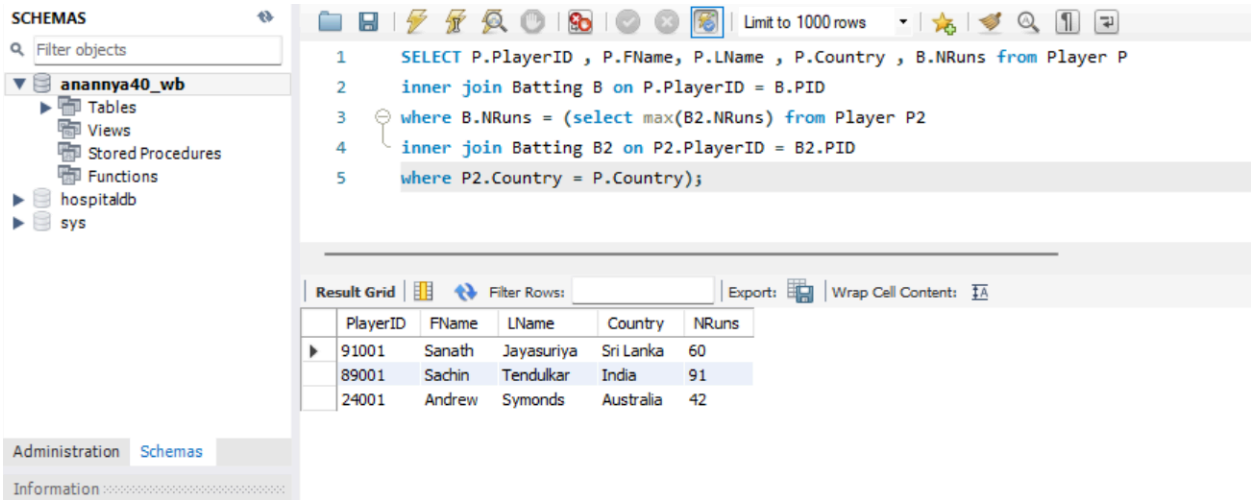
6. [Subquery - ALL Operator]: Write a query to find player batting performances where NRuns is greater than ALL runs scored by players representing 'Australia'.



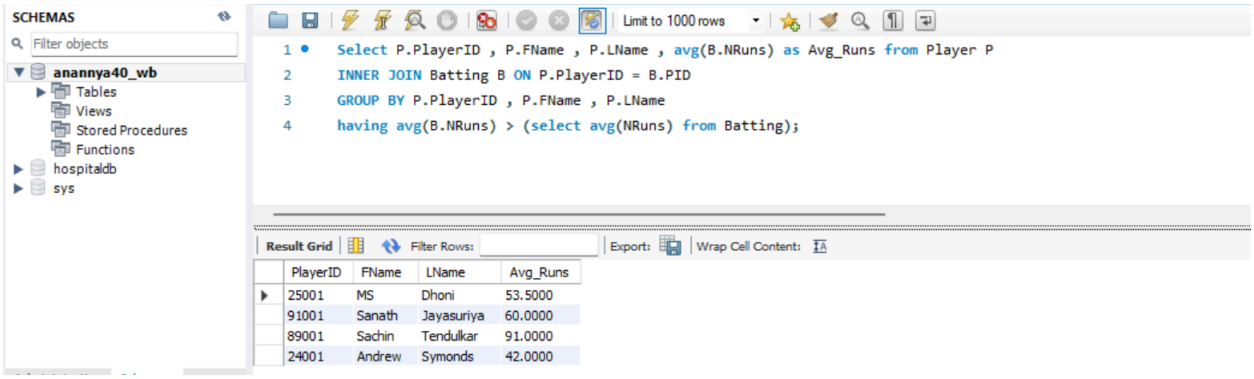
7. [Subquery - ANY / SOME]: Find all bowling performances where Nwickets is greater than ANY wicket count achieved by bowlers from 'Sri Lanka'.



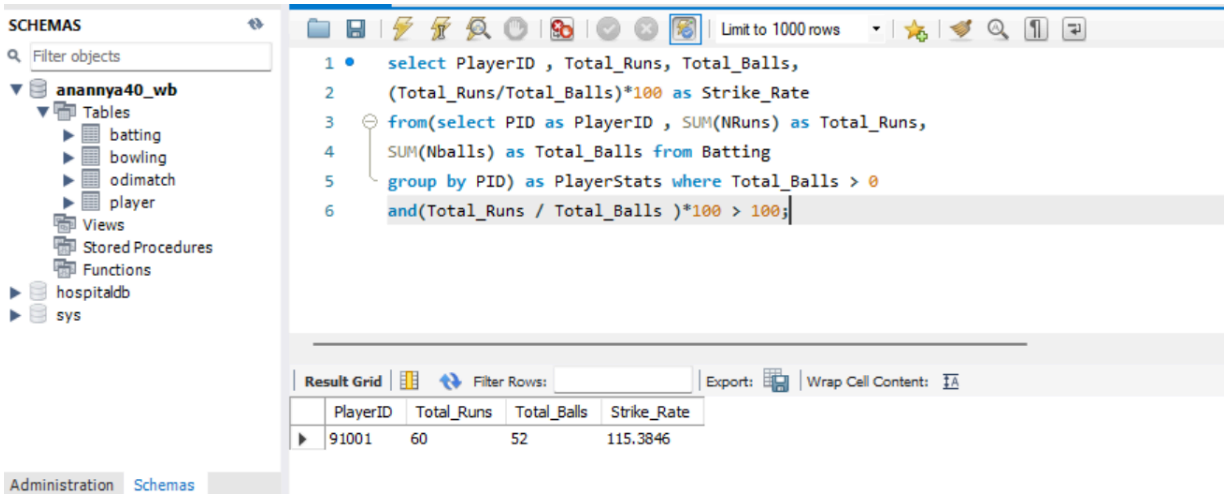
8 .[Correlated Subquery]: Write a correlated subquery to list each player who scored the maximum runs (NRuns) among all players from their own Country.



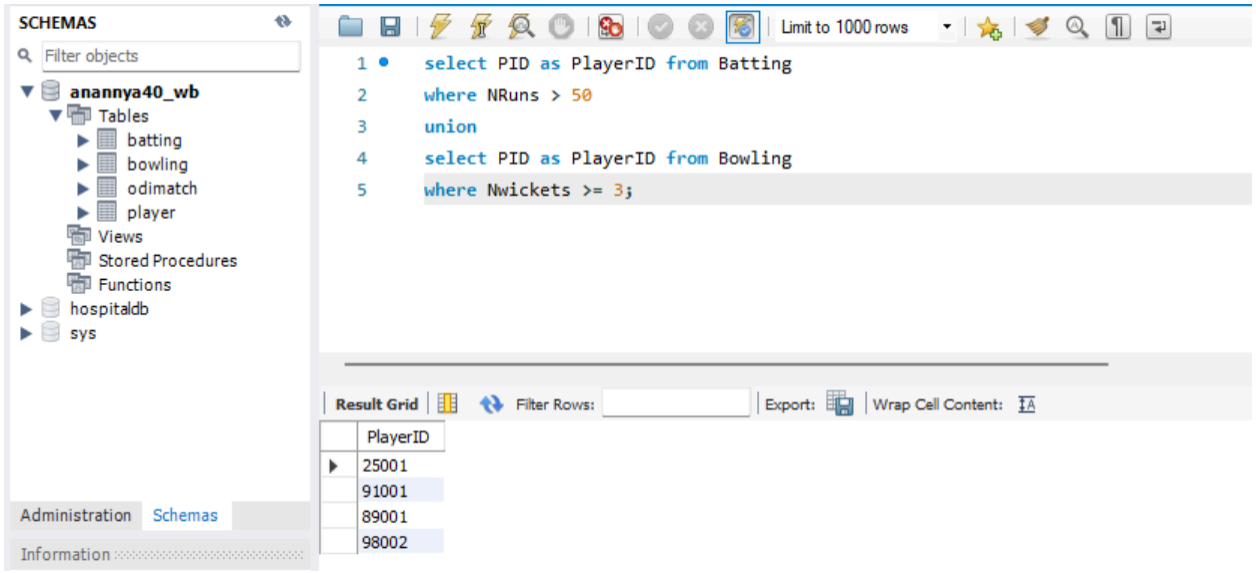
9. [Nested Subquery with HAVING]: Display players whose average runs per match is strictly greater than the overall average runs scored across all matches in the Batting table.



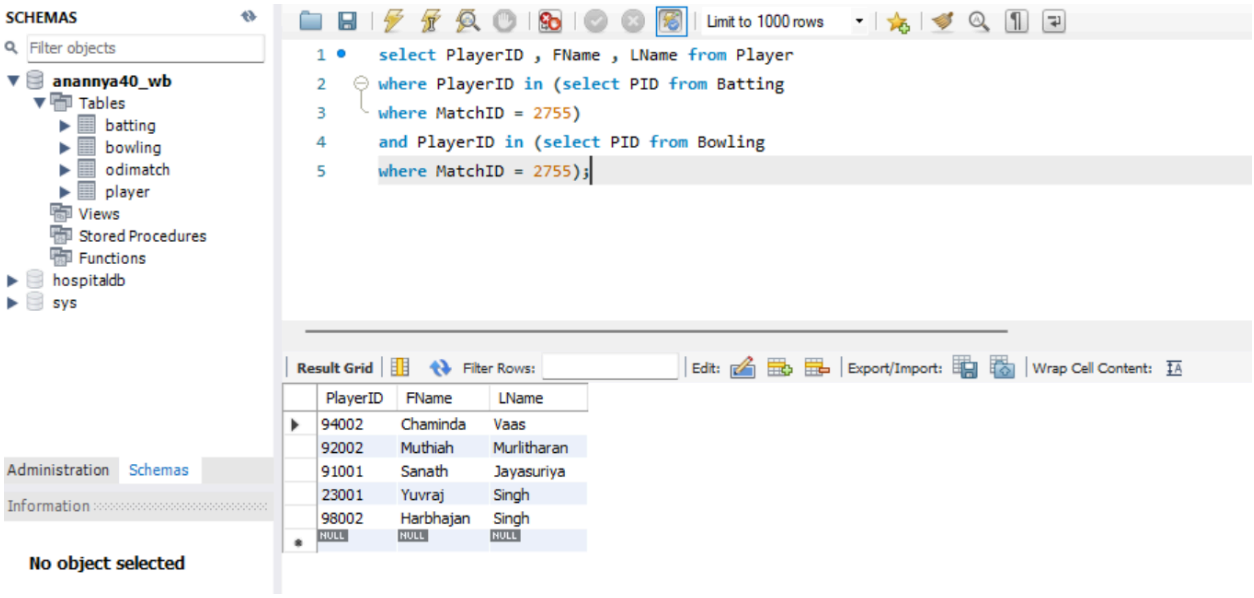
10. [Derived Table in FROM]: Create a derived table in the FROM clause calculating total runs and total balls faced per player, then select players with a strike rate ((Total_Runs / Total_Balls) * 100) greater than 100.



11. [Set Operator - UNION]: Write a query using UNION to combine the PlayerIDs of players who scored > 50 runs in Batting with players who took >= 3 wickets in Bowling.



12. [Set Operator / Subquery Intersect]: Find players who both batted AND bowled in MatchID = 2755 using an INTERSECT query or equivalent subquery structure.



13 .[Complex Grouping & HAVING]: Display Ground, total runs scored on that ground, and total matches played, considering only grounds where more than 2 matches took place and total runs exceed 100.

